

TRABAJO GRUPAL PIN2
Grupo 9
DevOps 2403

PIN2

Objetivos

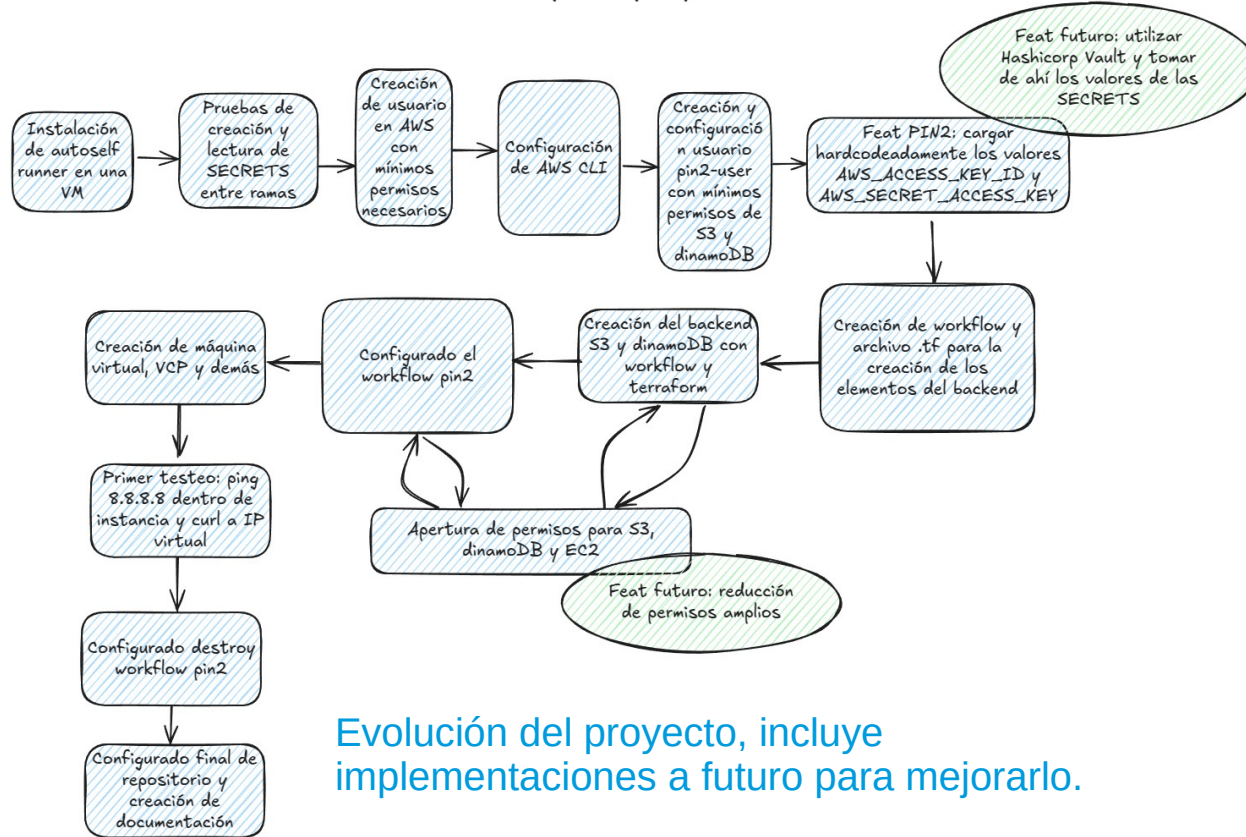
- * Objetivo principal: a través de GitHub Workflow y terraform desplegar una VPC, IGW, tablas de rutas, subred, SG y una instancia EC2, que configure con un script un servicio de Apache, utilizando como backend un bucket S3 y una tabla dinamoDB.
- * Objetivo secundario: a través de GitHub Workflow y terraform desplegar el backend.

Repositorio

https://github.com/MauroPereira/PIN2_public

PIN2

Roadmap del proyecto PIN2



Evolución del proyecto, incluye implementaciones a futuro para mejorarlo.

Existe dos GitHub workflows con sus archivos terraform separados por carpetas, “create_or_destroy_backend” para el objetivo secundario y “pin2” para el objetivo principal.

```
└─ .github / workflows
   ├── create_or_destroy_backed.yml
   └─ pin2.yml
└─ create_or_destroy_backed
   ├── create_or_destroy_backend.tf
   └─ doc_files
└─ pin2
   ├── all_code.txt
   ├── backend.tf
   ├── create_apache.sh
   ├── main.tf
   ├── setup.tf
   └─ polities
└─ README.md
```

PIN2 – create or destroy backend

create_or_destroy_backend.tf x

create_or_destroy_backed > create_or_destroy_backend.tf

```
1 terraform {
2   backend "local" {
3     path = "terraform.tfstate"
4   }
5 }
6
7 resource "aws_s3_bucket" "terraform_state_bucket" {
8   bucket = "mundose222-mauro-a-pereira"
9 }
10
11 resource "aws_dynamodb_table" "terraform_lock_table" {
12   name         = "terraformstatelock"
13   billing_mode = "PAY_PER_REQUEST"
14   hash_key     = "LockID"
15
16   attribute {
17     name = "LockID"
18     type = "S"
19   }
20
21   timeouts {
22     create = "5m"
23   }
24 }
25
```

Para crear el backend se utiliza un tfstate local, para eliminar los recursos al tratarse de un workflow y al ser efímero, se reconstruye el archivo tfstate y con esa información se destruyen los recursos.

```
- name: Importing Init
id: importing
run: |
  echo "Initializing Terraform..."
  terraform init

  echo "Importing existing S3 bucket..."
  terraform import aws_s3_bucket.terraform_state_bucket munde

  echo "Importing existing DynamoDB table..."
  terraform import aws_dynamodb_table.terraform_lock_table te

working-directory: create_or_destroy_backed
env:
  AWS_ACCESS_KEY_ID: ${ secrets.AWS_ACCESS_KEY_ID } Context
  AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
  AWS_DEFAULT_REGION: ${ secrets.AWS_DEFAULT_REGION }
continue-on-error: false

- name: Display Terraform State
id: display_tfstate
run: |
  echo "Displaying generated terraform.tfstate file:"
  cat terraform.tfstate

working-directory: create_or_destroy_backed
env:
  AWS_ACCESS_KEY_ID: ${ secrets.AWS_ACCESS_KEY_ID } Context
  AWS_SECRET_ACCESS_KEY: ${ secrets.AWS_SECRET_ACCESS_KEY }
  AWS_DEFAULT_REGION: ${ secrets.AWS_DEFAULT_REGION } Context
continue-on-error: false
```

PIN2 – create or destroy backend - GH Actions

← Create or destroy S3 and dinamoDB backend

✓ Create or destroy S3 and dinamoDB backend #10

Summary

Jobs

✓ terraform_create_backend

⌚ terraform_destroy_backend

Run details

⌚ Usage

📄 Workflow file

Manually triggered 2 days ago

👤 MauroPereira → f1c4183 [test](#)

Status

Success

Total duration

32s

Billable time

1m

Artifacts

-

create_or_destroy_backend.yml

on: workflow_dispatch

✓ terraform_create_backend 23s

⌚ terraform_destroy_backend 0s

PIN2 – create or destroy backend - GH Actions

Summary

Jobs

✓ terraform_create_backend

⌚ terraform_destroy_backend

Run details

⌚ Usage

📄 Workflow file

terraform_create_backend

succeeded 2 days ago in 23s

Search logs

✓ Terraform Apply

```
73 + logging (known after apply)
74
75 + object_lock_configuration (known after apply)
76
77 + replication_configuration (known after apply)
78
79 + server_side_encryption_configuration (known after apply)
80
81 + versioning (known after apply)
82
83 + website (known after apply)
84 }
85
86 Plan: 2 to add, 0 to change, 0 to destroy.
87 aws_dynamodb_table.terraform_lock_table: Creating...
88 aws_s3_bucket.terraform_state_bucket: Creating...
89 aws_s3_bucket.terraform_state_bucket: Creation complete after 1s [id=mundose222-mauro-a-pereira]
90 aws_dynamodb_table.terraform_lock_table: Creation complete after 7s [id=terraformstatelock]
91
92 Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
93 ::debug::Terraform exited with code 0.
```

> ✓ Post Checkout

> ✓ Complete job

PIN2 – pin2

Al haber creado el backed, este se utiliza para guardar la información de todos los recursos creados con pin2.

```
backend.tf x
pin2 > backend.tf
1 terraform {
2   backend "s3"{
3     bucket      = "mundose222-mauro-a-pereira"
4     # region    = "us-east-1" # No necesario
5     key         = "backend.tfstate"
6     dynamodb_table = "terraformstatelock"
7   }
8 }
9
```

Se ofrece una función adicional, se pueden crear los secrets necesarios para AWS en Actions.

```
workflow dispatch:
  inputs:
    action:
      description: "Action to perform"
      type: choice
      options:
        - Create backend
        - Destroy backend
        - Write AWS secrets and Create backend
  aws_access_key_id:
    description: "AWS Access Key ID"
    required: true
  aws_secret_access_key:
    description: "AWS Secret Access Key"
    required: true
  aws_default_region:
    description: "AWS Default Region"
    default: "us-east-1"
    required: false

jobs:
  write_aws_secrets:
    if: ${{ (github.event.name == 'workflow_dispatch' && github.event.inputs.action == 'Write AWS secrets and Create backend') }}
    name: "Write AWS secrets"
    runs-on: ubuntu-latest
    steps:
      - name: "Write AWS ACCESS KEY ID"
        uses: hmanzur/actions-set-secret@v2.0.0
        with:
          name: "AWS_ACCESS_KEY_ID"
          value: ${{ github.event.inputs.aws_access_key_id }}
          token: ${{ secrets.REPO_SECRET_WRITE_TOKEN }} Context access might be invalid: REPO_SECRET_WRITE_TOKEN

      - name: "Write AWS SECRET ACCESS KEY"
        uses: hmanzur/actions-set-secret@v2.0.0
        with:
          name: "AWS_SECRET_ACCESS_KEY"
          value: ${{ github.event.inputs.aws_secret_access_key }}
          token: ${{ secrets.REPO_SECRET_WRITE_TOKEN }} Context access might be invalid: REPO_SECRET_WRITE_TOKEN

      - name: "Write AWS DEFAULT REGION"
        uses: hmanzur/actions-set-secret@v2.0.0
        with:
          name: "AWS_DEFAULT_REGION"
          value: ${{ github.event.inputs.aws_default_region }}
          token: ${{ secrets.REPO_SECRET_WRITE_TOKEN }} Context access might be invalid: REPO_SECRET_WRITE_TOKEN
```

PIN2 – pin2 - GH Actions

← PIN2

✓ PIN2 #43

Summary

Jobs

- ✓ Write AWS secrets
- ✓ Read AWS secret
- ✓ terraform_backend_apply
- ⌚ terraform_backend_destroy

Run details

- ⌚ Usage
- 📄 Workflow file

Manually triggered yesterday

MauroPereira a9d07b2 **stable**

Status

Success

Total duration

1m 12s

Billable time

3m

pin2.yml

on: workflow_dispatch

✓ Write AWS secrets

6s

✓ Read AWS secret

0s

✓ terraform_backend_apply

47s

⌚ terraform_backend_destroy

0s

PIN2 – pin2 - GH Actions

Summary

Jobs

- ✓ Write AWS secrets
- ✓ Read AWS secret
- ✓ **terraform_backend_apply**
- ⌚ terraform_backend_destroy

Run details

- ⌚ Usage
- 📄 Workflow file

terraform_backend_apply

succeeded yesterday in 47s

Search logs

✓ Terraform Apply

```
250 + tags_all = {
251 + "Name" = "terraform-vpc"
252 }
253 }
254
255 Plan: 6 to add, 0 to change, 0 to destroy.
256
257 Changes to Outputs:
258 + Webserver-Public-IP = (known after apply)
259 aws_vpc.vpc: Creating...
260 aws_vpc.vpc: Still creating... [10s elapsed]
261 aws_vpc.vpc: Creation complete after 13s [id=vpc-07c6c43a4ae74ab2d]
262 data.aws_route_table.main_route_table: Reading...
263 aws_internet_gateway.igw: Creating...
264 aws_subnet.subnet: Creating...
265 aws_security_group.sg: Creating...
266 data.aws_route_table.main_route_table: Read complete after 0s [id=rtb-0f06ce5adae02a5d1]
267 aws_internet_gateway.igw: Creation complete after 0s [id=igw-05432f1370f7526a1]
268 aws_default_route_table.internet_route: Creating...
269 aws_subnet.subnet: Creation complete after 1s [id=subnet-00e398bc95108b987]
270 aws_default_route_table.internet_route: Creation complete after 2s [id=rtb-0f06ce5adae02a5d1]
271 aws_security_group.sg: Creation complete after 3s [id=sg-03f3dc3535ae11a52]
272 aws_instance.webserver: Creating...
273 aws_instance.webserver: Still creating... [10s elapsed]
274 aws_instance.webserver: Creation complete after 13s [id=i-0a0a4f1907eb618dc]
275
276 Apply complete! Resources: 6 added, 0 changed, 0 destroyed.
```

Chequeo final

vpc-07c6c43a4ae74ab2d / terraform-vpc Acciones ▼

Detalles Información

ID de la VPC vpc-07c6c43a4ae74ab2d	Estado ✓ Available	Bloquear el acceso público Desactivado	Nombres de host de DNS Habilitado
Resolución de DNS Habilitado	Tenencia default	Conjunto de opciones de DHCP dopt-0767fd223619f7c86	Tabla de enrutamiento principal rtb-0f06ce5adae02a5d1 / Terraform-RouteTable
ACL de red principal acl-01a323885e3aa8882	VPC predeterminada No	CIDR IPv4 10.0.0.0/16	Grupo IPv6 -
CIDR IPv6 (grupo de bordes de red) -	Métricas de uso de direcciones de red Desactivado	Grupos de reglas del firewall de DNS de Route 53 Resolver -	ID de propietario 593793067544

Mapa de recursos CIDR Registros de flujo Etiquetas Integraciones

Mapa de recursos Información

VPC [Mostrar detalles](#)

Su red virtual de AWS

terraform-vpc

Subredes (1)

Subredes dentro de esta VPC

us-east-1a

subnet-00e398bc95108b987

Tablas de enrutamiento (1)

Dirigir el tráfico de red a los recursos

Terraform-RouteTable

Conexiones de red (1)

Conexiones a otras redes

igw-05432f1370f7526a1

Se pueden observar todos los recursos creados en la VPC.

Chequeo final

```
ec2-user@ip-10-0-1-15 ~]$ ping -c 3 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=117 time=0.658 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=117 time=1.23 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=117 time=0.720 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2061ms
rtt min/avg/max/mdev = 0.658/0.870/1.233/0.257 ms
ec2-user@ip-10-0-1-15 ~]$
```

i-0a0a4f1907eb618dc (webserver)
PublicIPs: 3.95.205.250 PrivateIPs: 10.0.1.15

Desde adentro de la instancia EC2 efectivamente hay conexión a internet, lo que demuestra que el despliegue está correcto.

Desde una terminal cualquiera, fuera de la nube privada de AWS, se puede obtener la información de Apache.



```
mauro@map-xu22-z170xps11:~$ curl http://3.95.205.250/
<!DOCTYPE html>
<html lang="es">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Página de Inicio - PIN2 DevOps 2403</title>
</head>
<body>
  <h1>PIN2 DevOps 2024</h1>
  <p><a href="https://github.com/MauroPereira/PIN2" target="_blank">Project Repository</a></p>
  <p>Created by
    <a href="mailto:mauro.a.p.pereira@gmail.com">Pereira, Mauro Alejandro</a>
  </p>
</body>
</html>
```